

2019-2020-1 学期《线性代数与解析几何》A 卷参考答案

一、填空题：共 6 题，每题 3 分，共 18 分。

1. 20 2. $\begin{pmatrix} 6 & 0 & 0 \\ -18 & 3 & 0 \\ 12 & -6 & 2 \end{pmatrix}$ 3. 2 4. $x^2 + y^2 = 4z$ 5. 3, 1, 2 6. $\frac{4}{5}$

二、选择题：共 6 题，每题 3 分，共 18 分。

1. D 2. B 3. A 4. B 5. A 6. C.

三、(8分)

$$D = \begin{vmatrix} 4 & 3 & 3 & \cdots & 3 & 3 \\ -1 & 2 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & 0 & 3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & 0 & 0 & \cdots & n-1 & 0 \\ -1 & 0 & 0 & \cdots & 0 & n \end{vmatrix} \begin{matrix} c_1 + \frac{1}{3}c_2 \\ c_1 + \frac{1}{3}c_3 \\ \vdots \\ c_1 + \frac{1}{n}c_n \\ = \end{matrix} \begin{vmatrix} 4 + \sum_{i=2}^n \frac{3}{i} & 3 & 3 & \cdots & 3 & 3 \\ 0 & 2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & n-1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & n \end{vmatrix} = \left(4 + 3 \sum_{i=2}^n \frac{1}{i}\right) \cdot n! = \left(1 + 3 \sum_{i=1}^n \frac{1}{i}\right) \cdot n!$$

四、(共 2 小题, 每小题 5 分, 共 10 分)

(1) l_2 方向向量= $\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & -1 \\ 2 & -1 & 4 \end{vmatrix} = 7\vec{i} - 14\vec{j} - 7\vec{k}$ (+3分)

$$l_1 \text{ 方向向量 } \vec{s} = \vec{i} - 2\vec{j} - \vec{k}$$

l_1 的对称式方程: $\frac{x-1}{1} = \frac{y-2}{-2} = \frac{z-3}{-1}$ (+2分)

(2) 设所求平面方程: $2x-2y-z+D=0$

点 $P(0,0,D)$ 是平面上的点.

点 P 到平面 $2x-2y-z+D=0$ 的距离 $= \frac{|2 \times 0 - 2 \times 0 - 1 \times D + 3|}{\sqrt{2^2 + (-2)^2 + (-1)^2}} = 3$ (+3分)

$$|-D+3|=9, \text{ 得 } D=12 \text{ 或 } D=-6$$

所求平面方程: $2x-2y-z+12=0$ 或 $2x-2y-z-6=0$ (+2分)

五、(15分)

$$\begin{aligned} \tilde{A} = \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & a \\ 3 & 2 & 1 & 1 & -3 & 0 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 5 & 4 & 3 & 3 & -b & 2 \end{array} \right) &\longrightarrow \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & a \\ 0 & -1 & -2 & -2 & -6 & -3a \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & -1 & -2 & -2 & -5-b & 2-5a \end{array} \right) \\ &\longrightarrow \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & a \\ 0 & 0 & 0 & 0 & 0 & 3-3a \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 1-b & 5-5a \end{array} \right) &\longrightarrow \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & a \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 1-b & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-a \end{array} \right) \quad (+4\text{分}) \end{aligned}$$

(1) 当 $a \neq 1$ 时, $r(A) < r(\tilde{A})$, 方程组无解 (+3分)

当 $a=1, b=1$ 时, $r(A)=r(\tilde{A})=2$, 方程组有无穷多解

$$\tilde{A} \longrightarrow \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \begin{cases} x_1 + x_2 + x_3 + x_4 + x_5 = 1 \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 3 \end{cases} \Rightarrow \begin{cases} x_1 = k_1 + k_2 + 5k_3 - 2 \\ x_2 = -2k_1 - 2k_2 - 6k_3 + 3 \\ x_3 = k_1 \\ x_4 = k_2 \\ x_5 = k_3 \end{cases}$$

$$\text{通解: } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = k_1 \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + k_3 \begin{pmatrix} 5 \\ -6 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{其中, } k_1, k_2 \text{ 是任意常数} \quad (+4\text{分})$$

当 $a=1, b \neq 1$ 时, $r(A)=r(\tilde{A})=3$, 方程组有无穷多解

$$\tilde{A} \longrightarrow \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 1-b & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \begin{cases} x_1 + x_2 + x_3 + x_4 + x_5 = 1 \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 3 \\ (1-b)x_5 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = k_1 + k_2 - 2 \\ x_2 = -2k_1 - 2k_2 + 3 \\ x_3 = k_1 \\ x_4 = k_2 \\ x_5 = 0 \end{cases}$$

$$\text{通解: } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = k_1 \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{其中, } k_1, k_2 \text{ 是任意常数} \quad (+4\text{分})$$

六、(10 分)

解: (1)

$$(\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 2 & 6 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 7 & -19 & -6 \\ -2 & 5 & 3 \\ -5 & 14 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 6 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -3 & 2 & 0 \end{pmatrix}$$

$$\beta_1 = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}, \quad \beta_2 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \quad \beta_3 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad +6 \text{ 分}$$

$$(2) \quad \eta \text{ 在基 } \alpha_1, \alpha_2, \alpha_3 \text{ 下的坐标 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 6 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \quad +4 \text{ 分}$$

七、(15 分)

解 (1)

$$|\lambda E - A| = \begin{vmatrix} \lambda+1 & -2 & -2 \\ -2 & \lambda+1 & 2 \\ -2 & 2 & \lambda+1 \end{vmatrix} = (\lambda-1)^2 (\lambda+5)$$

A 的全部特征值: $\lambda_1 = \lambda_2 = 1, \lambda_3 = -5$ (+5分)

(2)

对于 $\lambda=1$, 解方程组 $(E-A)X=0$

$$E-A = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & 2 \\ -2 & 2 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{得到对应 } \lambda=1 \text{ 的特征向量: } \alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \alpha_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad (+3 \text{ 分})$$

对于 $\lambda=-5$, 解方程组 $(-5E-A)X=0$

$$-5E-A = \begin{pmatrix} -4 & -2 & -2 \\ -2 & -4 & 2 \\ -2 & 2 & -4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{得到对应 } \lambda=-5 \text{ 的特征向量: } \alpha_3 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \quad (+2 \text{ 分})$$

正交化:

$$\beta_1 = \alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \beta_2 = \alpha_2 - \frac{(\beta_1, \alpha_2)}{(\beta_1, \beta_1)} \beta_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix}$$

单位化:

$$\eta_1 = \frac{1}{|\beta_1|} \beta_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \eta_2 = \frac{1}{|\beta_2|} \beta_2 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}, \quad \eta_3 = \frac{1}{|\alpha_3|} \alpha_3 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\text{得正交矩阵: } T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \quad T^{-1}AT = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{pmatrix} \quad (+5\text{分})$$

八、(6分)

证明 设 λ 是 A 的任一特征值.

而 $A^3 - 4A^2 + 5A - 2E$ 的特征值是 $\lambda^3 - 4\lambda^2 + 5\lambda - 2$

由于 $A^3 - 4A^2 + 5A - 2E = 0$, 则 $\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$

$(\lambda - 1)^2(\lambda - 2) = 0$, $\lambda = 1$ 或 $\lambda = 2$

A 的任何特征值只能为1或2, 所以 A 的全体特征值都大于0,

由于 A 是实对称矩阵, 因此 A 是正定矩阵.